SQL & MySQL Practice Exercises

Comprehensive Workbook — 26 Questions Across Core Database Concepts

Topic: SQL / MySQL Relational Databases **Questions:** 26 Exercises **Format:** Student Practice Sheet

Section 1: Basic Selection, Filtering, and Sorting

1. Write an SQL query to retrieve all columns for every record in the `customers` table.

2. Retrieve the `product_name`, `category`, and `price` for all items in the `products` table, sorted by `price` from highest to lowest.

3. Write a query to find all customers whose `city` is listed as either 'New York' or 'Chicago'.

4. Retrieve all products that have a stock quantity greater than 0 and a price less than $100.

5. Write a query to list all customers whose `email` address ends with `@gmail.com`.

6. Select all orders placed between '2026-01-01' and '2026-06-30' from the `orders` table.

7. Write a query to return the top 5 most expensive products in the `products` table.

Section 2: Aggregations and Grouping

8. Calculate the total number of registered customers in the `customers` table.

9. Find the average price of products grouped by `category`.

10. Write a query to find the minimum and maximum order total amounts from the `orders` table.

11. Count how many orders were placed by each `customer_id`.

12. Retrieve product categories that have an average price greater than $250 using the `HAVING` clause.

13. Calculate the total revenue generated across all completed orders.

Section 3: Joins and Multi-Table Queries

14. Write an `INNER JOIN` query to display `order_id`, `order_date`, and the corresponding customer's `first_name` and `last_name`.

15. Perform a `LEFT JOIN` between `customers` and `orders` to list all customers along with their order details, including customers who have never placed an order.

16. Retrieve a list of product names and their associated order quantities by joining `products` and `order_items`.

17. Write a query to find any categories that currently have no products assigned to them.

18. Display `order_id`, customer full name (`first_name` and `last_name`), `product_name`, and `quantity` by joining `customers`, `orders`, `order_items`, and `products`.

Section 4: Subqueries and Conditional Logic

19. Write a query to find all products whose price is strictly higher than the overall average price of all products in the database.

20. Retrieve all customers who have placed at least one order with a total amount exceeding $1,000 using an `IN` or `EXISTS` subquery.

21. Write an SQL query using a `CASE` expression to label products as 'In Stock' if `stock_quantity` > 0 and 'Out of Stock' if `stock_quantity` = 0.

22. Find the `customer_id` of the user who has spent the highest cumulative amount across all orders.

Section 5: Data Definition & Data Manipulation (DDL / DML)

23. Write an `INSERT INTO` statement to add a new product into the `products` table with values for `product_name`, `category`, `price`, and `stock_quantity`.

24. Write an `UPDATE` statement to apply a 10% price reduction to all items belonging to the 'Electronics' category.

25. Write a `DELETE` statement to remove records from the `orders` table where the status is 'Cancelled' and the order date is prior to '2025-01-01'.

26. Write a `CREATE TABLE` SQL statement for an `audit_log` table containing `log_id` (Primary Key, Auto-Increment), `user_id` (Integer), `action_performed` (VARCHAR 255), and `created_at` (TIMESTAMP/DATETIME).

Instructions for Students: Write your SQL queries in standard MySQL syntax. Make sure to test your queries against the sample schema provided in class or in the reference guidebook.